CopilotTraining
Technical briefing

HydraFusion

How It Works and How to Use It

Patterns, reasoning effort, and practical use

GitHub Copilot · 20 minutes · Research preview · October 26, 2026

🤔 The Core Question

"How does HydraFusion orchestrate models, and how do we use it?"

Understand the runtime, enable it in Copilot, and try representative prompts.
Read the execution and verify the result.

Understand orchestration

Compare Auto, HydraFusion and Max; trace the execution paths.

Use it in Copilot

Enable the preview and try explanation, implementation or analysis.

Read the outcome

Inspect actual progress, changes and per-model usage.

3 patterns

Single, Cascade and Critique; chosen for each prompt.

3 clients

CLI, VS Code and the Copilot app.

Research preview

No SLA; not intended for production workloads.

Table of Contents

Part 1

Understand the orchestrator

Auto, HydraFusion and Max

Separate model routing from workflow orchestration.

Part 2

Follow the execution

Single, Cascade and Critique

Trace selective solving and review passes.

Part 3

Use HydraFusion

Enable, select and prompt

Use exact commands and representative examples.

Part 4

Evaluate the result

Progress, changes and usage

Read what ran and what still needs verification.

Part 1

Understand the orchestrator

A model-picker entry delegates workflow, models and reasoning effort.

Auto
Route the task to one model.

HydraFusion
Select a workflow for the prompt.

Max
Apply depth-oriented orchestration.

```
HydraFusion is a runtime orchestrator.
↳ Extra passes are selective; one solver can be enough.
```

Auto routes a model; HydraFusion selects a workflow

➡ Auto · one-model routing

Input: a normal prompt
Auto considers task fit and service health.

System decision
Select one model to handle the task.

Useful baseline
Routine explanations and focused coding work.

🔀 HydraFusion · runtime orchestration

Input: a normal prompt
The orchestrator selects how to solve it.

System decisions
Execution pattern, participating models and stage effort.

Selective passes
One solver, conditional escalation or independent review.

💡 HydraFusion is a system behind a picker entry, not a new foundation model.

Max tunes the same architecture for deeper reasoning

⚖️ HydraFusion

Balanced execution
Coordinate solving and review with token efficiency in mind.

Stage-specific reasoning
Drafting uses medium; reviews use high effort.

Starting point
Well-scoped coding work where review may help.

🧠 HydraFusion Max

Depth-oriented tuning
Use the same workflow architecture for demanding tasks.

Higher effort where useful
Max can use extra-high reasoning.

When to evaluate it
Complex analysis where thoroughness justifies more execution.

🔀 The system chooses models and effort. Users cannot force a pattern or constituent model.

Part 2

Follow the execution

The system chooses a path for each prompt and returns one final response.

➡️

Single

One solving pass.

⚖️

Cascade

Draft, judge, then further work if needed.

🔍

Critique

Independent review and solver revision.

```
The following paths are conceptual architecture.
↳ A follow-up can select a different path.
```

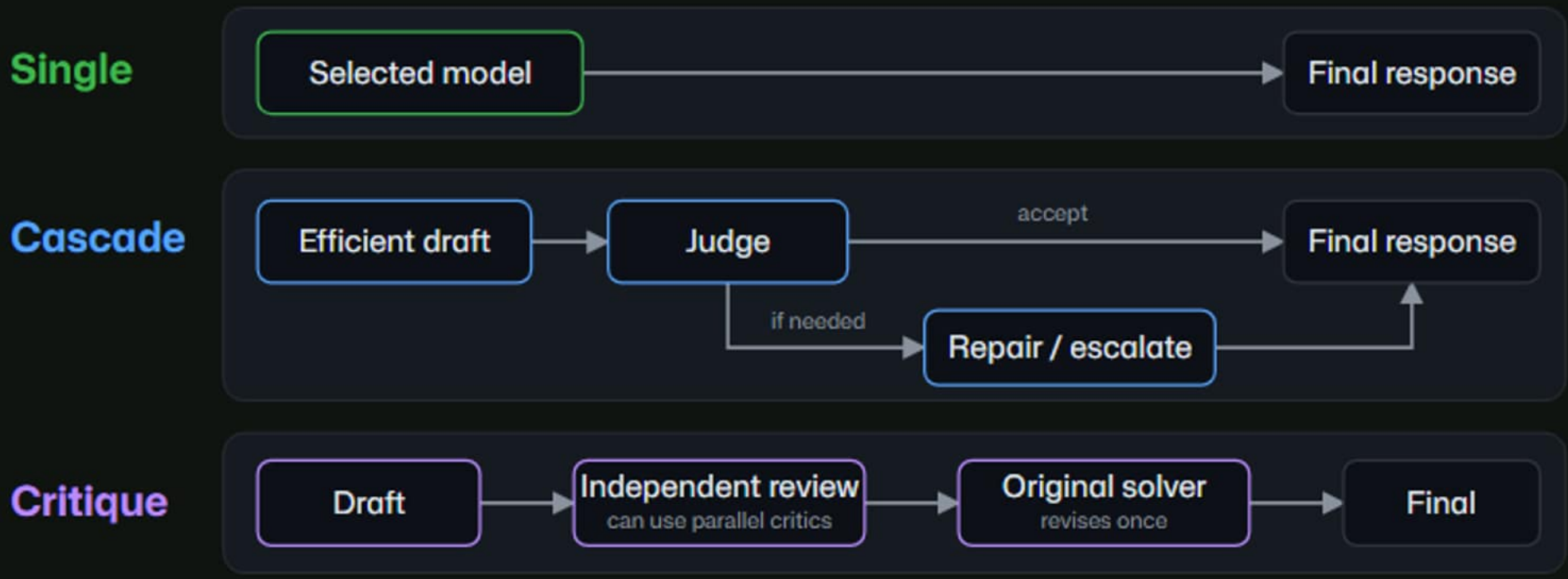



Execution paths

1 of 2

Three paths to one final response

HydraFusion chooses a pattern for each prompt. These paths are conceptual.



Adapted from GitHub Docs, "Using HydraFusion," and the October 14 update. Extra model passes are selective.

Judges and critics inspect work at review depth

⚖️

Cascade judge and repair

The judge evaluates the draft and can request repair or escalation when needed.

📁

Workspace-aware critics

Critics use tools to inspect actual files. Parallel critics can review the work.

🧠

Reasoning per stage

Medium for drafting, high for review; Max can use extra-high reasoning.

🔀

Separate from subagents

HydraFusion coordinates its internal passes. It does not start subagents.

Context display is conservative across constituent models; HydraFusion has no independent context window.

Part 3

Use HydraFusion

Enable the preview, select an option, and send a normal Copilot prompt.



Enable

CLI, VS Code or Copilot app.



Select

HydraFusion or HydraFusion Max.



Prompt

Explain, implement or analyze.

```
Send a normal Copilot prompt.
↳ Inspect the path, response and workspace changes.
```

Enable HydraFusion, then select it in the picker



Copilot CLI

Start copilot --experimental.

- Use /model.
- Select HydraFusion or HydraFusion Max (Research Preview).



VS Code

Use version 1.140+ or Insiders.

- Enable chat.copilot.hydraFusion.enabled.
- Select HydraFusion or Max in Chat.



Copilot app

Use the current app.

- Settings → Experimental → HydraFusion.
- Select HydraFusion or Max.

🛡️ Availability depends on plan, preview access and allowed-model policy. Check settings and administrator access.

Start, select, prompt, then inspect progress

Interactive CLI workflow

```
copilot --experimental

# Inside the session:
/model
# Select HydraFusion or HydraFusion Max

# Send a normal prompt; watch progress.
/usage
```

Visible execution

Watch actual pattern, planned passes, progress and elapsed time.

Per-model usage

/usage reports session consumption. Record a starting point for prompt comparisons.

Optional live call

Cap execution at two minutes. Then inspect or disclose what remains unverified.

Try explanation, implementation or demanding analysis



Explain

Explain this function and list its boundary cases. Do not edit files.

Inspect: explanation against actual code.

Observe: the selected pattern.



Implement

Add an optional timeout; preserve callers, add edge-case tests and run them.

Inspect: changed files and regressions.

Verify: actual test results.



Analyze

Compare two recovery designs against these failure scenarios. State assumptions and tradeoffs.

Inspect: coverage and unresolved questions.

Consider Max when depth matters.

Prompts and inspection goals, not observed outputs or routing guarantees. Single remains a valid outcome.

Part 4

Evaluate the result

Use actual progress, file changes and usage to understand what ran.



Progress

Inspect actual pattern and passes.



Result

Verify behavior and workspace changes.



Usage

Read participating-model consumption.

```
A review pass contributes analysis.
↳ Correctness still needs independent checks.
```

Inspect what ran and whether the result fits

Actual execution

Read the selected pattern, planned passes, progress, elapsed time and warnings.

Response and models

Inspect the final response. In VS Code or the app, hover the response for model details.

Workspace and behavior

Read changed files, verify relevant behavior and independently rerun appropriate checks.

Session usage

Use /usage for per-model session consumption. Record a starting point for comparisons.

If a result is wrong: give one precise failing example, narrow the task, and rerun relevant checks.

Keep workspace changes, billing and preview limits visible

Draft edits can persist

Discarding a draft does not undo edits automatically. Inspect files after stops or failures.

Per-model billing

Each model uses its standard rate. No orchestration surcharge; no Auto discount.

Access and permissions

Administrator policy governs models and preview access. Normal tool permissions stay in force.

Research preview

HydraFusion and Max have no SLA and are not intended for production workloads.

Historical offline results show quality/cost tradeoffs; they do not forecast Max or local savings.

The Value Recap

From a picker entry to an inspectable runtime

Before

- Identify the selected option
- Read the final response
- Inspect changed code
- Estimate execution cost

After

- Distinguish routing from orchestration
- Read the actual pattern and passes
- Verify behavior and workspace state
- Inspect per-model session usage

3
conceptual execution patterns

3
supported client surfaces

1
final response per prompt

✅ What You Can Do Today

Today

- Enable the preview in one client.
- Try a read-only explanation.
- Observe actual progress and models.

This Week

- Try one bounded non-production edit.
- Inspect files and rerun relevant checks.
- Record real elapsed time and usage.

This Month

- Compare identical inputs across sessions.
- Evaluate Max on demanding analysis.
- Recheck changing preview guidance.

🔑 Key Takeaway

Understand the workflow, use normal Copilot prompts, and verify the actual result.

References

Current operating guidance

Using HydraFusion
Setup, patterns, progress, billing and limitations.

Auto model selection
Routing behavior and Auto tiers.

Mechanism and dated evidence

October 14 update and Max
Stage effort, judge/repair and workspace-aware critics.

Project HydraFusion
Architecture and historical offline evaluations.

Understand. Follow. Use. Evaluate.

HydraFusion: How It Works and How to Use It

Workflow
Selected for each prompt.

Use
Enable, select and prompt.

Evidence
Inspect progress, results and usage.

Which existing task would make a useful comparison because its result is easy to verify?